

Toward a Rotation-Independent Magnetar Formation Channel in NVG: Dense-Matter Electromagnetic Response, Limits, and Observational Tests

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Repository: <https://github.com/infosave2007/vmf>

Abstract

The standard magnetar scenario attributes magnetic fields of $B \sim 10^{14} - 10^{15}$ G to an early-time α - ω dynamo, which in turn requires proto-neutron-star birth periods of order $P_i \sim 1 - 3$ ms. This requirement is in tension with several observational facts, including the absence of clear evidence for extreme rotational energy injection in some associated supernova remnants and the existence of ultra-long-period magnetar candidates such as 1E 161348-5055. In this note we reformulate the Null-Vector Gravity / Vacuum Mass Fraction (NVG/VMF) magnetar idea in a more conservative and testable way.

The robust part of the framework is the dense-matter sector: the same NVG/VMF construction used elsewhere in this repository links the QCD vacuum anchor $M_{\Omega,0} \approx 859$ MeV to neutron-star equation-of-state calculations capable of reaching $M_{\max} \approx 2.2 - 2.3 M_{\odot}$ with $R_{1.4} \approx 12$ km. The less settled step is the electromagnetic one: a dense-matter modification of the effective electromagnetic response can be motivated through a vacuum/order-parameter-dependent gauge kinetic term, but a first-principles derivation of an effective magnetic permeability law for neutron-star cores is not yet complete.

Accordingly, we present the magnetar mechanism not as a finished proof, but as a quantitative working hypothesis: if the same in-medium structural transition that modifies the EOS also changes the local electromagnetic constitutive response, then a pre-existing fossil field can be amplified during collapse without requiring millisecond birth rotation. We identify the three decisive next tests: deriving the effective electromagnetic response from more standard microphysics, verifying joint consistency with neutron-star masses/spins/supernova-remnant energetics, and confronting the model with the observed magnetar population, including long-period systems.

1. Introduction

Magnetars are neutron stars with inferred surface dipole fields typically in the range $10^{14} - 10^{15}$ G. In the canonical picture, these fields are generated by a turbulent dynamo during the first seconds of proto-neutron-star evolution. The difficulty is well known: the dynamo is efficient only if the star is born with a spin period P_i of a few milliseconds. That requirement becomes problematic when one considers systems for which the observational record does not favor such a rapidly rotating birth state.

This tension motivates alternative channels in which strong magnetic fields arise from internal structural physics rather than from extreme rotational energy. The purpose of the present version is not to claim that the problem is solved, but to sharpen a specific NVG/VMF possibility into a form that is easier to falsify.

The central idea is simple. In the NVG/VMF program, dense matter is not treated as moving through an inert vacuum with immutable properties. Instead, the nonperturbative QCD contribution to hadronic mass is represented by a macroscopic order parameter whose in-medium evolution affects the equation of state. If the same structural sector also contributes to the electromagnetic response of dense matter, then a collapse-induced change in that response may amplify a pre-existing magnetic field even for a slowly rotating newborn neutron star.

2. What Is Already on Firmest Ground in the Project

The magnetar proposal should be anchored first to the parts of the NVG/VMF framework that are already numerically developed.

In the dense-matter sector, the repository consistently uses a vacuum nonperturbative mass scale

$$M_{\Omega,0} \approx 859 \text{ MeV},$$

obtained from the nucleon mass budget after subtracting lattice-motivated sigma-term contributions. In the phenomenological EOS construction, this vacuum contribution decreases with baryon density through an ansatz of the form

$$M_{\Omega}(n_B) = M_{\Omega,0} \left(1 + \kappa_2 \frac{n_B}{n_0} \right)^{-\kappa_1/\kappa_2},$$

with calibrated benchmark values near $\kappa_1 = 0.25$ and $\kappa_2 = 0.80$.

Within the repository, this dense-matter chain is used to construct neutron-star equations of state and TOV solutions that yield, at benchmark level, realistic stellar properties: maximum masses around $2.2 - 2.3 M_{\odot}$, radii near $R_{1.4} \approx 12$ km, and acceptable compatibility with tidal deformability and basic spin limits. That part matters because any magnetar mechanism proposed inside NVG must

survive exactly the same dense-matter constraints. A magnetic-field story that works only after breaking the EOS sector would not be credible.

Therefore the magnetar paper should be read as a branch of the existing dense-matter program, not as a standalone electromagnetic speculation.

3. The Missing Bridge: From NVG Dense Matter to Electromagnetic Response

The weakest point of the original magnetar draft is also the most important one: the step from vacuum-mass melting to a specific law for effective magnetic permeability.

At the level of effective field theory, the basic route is clear. If the macroscopic order parameter $\mathcal{W}$ controls part of the medium structure, then the electromagnetic sector can couple through a gauge-kinetic function,

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}Z_{\text{EM}}(\mathcal{W})F_{\mu\nu}F^{\mu\nu}.$$

When $\mathcal{W}$ takes its vacuum value, one recovers standard low-density electrodynamics. When $\mathcal{W}$ shifts in dense matter, the constitutive response changes. In that sense, an effective electromagnetic response in dense matter is a natural EFT possibility.

However, one must separate three distinct statements.

First, it is reasonable to expect some in-medium electromagnetic modification once the charged quark sector and vacuum polarization are altered.

Second, the repository already contains partial formal work in this direction: dense matter can be associated with a modified effective dielectric response through quark-loop arguments and an order-parameter-dependent EM sector.

Third, and crucially, this does **not** yet amount to a first-principles derivation of a neutron-star-core magnetic permeability collapse strong enough to guarantee a large amplification factor. The current project materials themselves explicitly note that ϵ_0 and μ_0 cannot be derived from $M_{\Omega,0}$ alone, because leptonic contributions and the UV electroweak sector are not fixed by the QCD anchor. That limitation should be stated in the magnetar paper, not hidden.

For this reason, the correct status of the magnetic-response step is:

1. The existence of some dense-matter electromagnetic modification is plausible within the NVG/VMF EFT logic.
2. A direct bridge from NVG order-parameter dynamics to standard many-body/QCD-matter response functions is still needed.
3. Any large magnetic amplification factor should presently be treated as a benchmark hypothesis to be derived or rejected, not as an already closed theorem.

4. A Conservative Magnetar Scenario Inside NVG

With that caveat in place, the structural magnetar idea can be stated cleanly.

Suppose a collapsing stellar core carries a fossil magnetic field. Standard flux freezing already amplifies that field substantially as the core contracts. Now suppose, in addition, that the same dense-matter transition responsible for the NVG/VMF EOS also changes the local electromagnetic constitutive response in the inner core. Then the total post-collapse field may be written schematically as

$$B_{\text{NS}} \sim \Gamma_{\text{flux}} \Gamma_{\text{struct}} B_{\text{prog}},$$

where Γ_{flux} is the usual geometric compression factor and Γ_{struct} is a density-driven structural amplification factor.

The original preprint effectively assumed that Γ_{struct} can be of order several, with a benchmark volume-averaged value near 7.4. In the present version, that number should be interpreted more carefully: it is a target scale suggested by the earlier implementation, not yet a uniquely established prediction of standard microphysics.

Even in this more cautious form, the idea remains astrophysically interesting. If Γ_{struct} is indeed appreciable, then the need for millisecond birth periods is weakened or removed. In that case, highly magnetized neutron stars could be produced from a combination of reasonably strong progenitor fossil fields and density-triggered internal amplification, rather than from extreme birth rotation.

This is exactly the qualitative regime relevant to the slow-rotating magnetar paradox.

5. Quantitative Closure Checks

To reduce the most obvious ambiguities, the repository now includes a dedicated script, `verification/nvg_magnetar_closure.py`, which collects the disputed pieces into one calculation chain: dense-matter electromagnetic response, flux-freezing plus structural amplification, rotational-energy bookkeeping, and a dipole spin-down check for a 1E 161348-5055-like system.

Three immediate numerical results matter.

First, the dense-matter electromagnetic bridge is nonzero but not by itself decisive. Using the same VMF melting law, the simplified quark-loop estimate gives an effective dielectric ratio in the approximate range

$$\epsilon_{\text{eff}}/\epsilon_0 \approx 0.93 \ (n_B = n_0), \quad \epsilon_{\text{eff}}/\epsilon_0 \approx 0.83 \ (n_B = 5n_0).$$

So the sign of the effect is not the problem: dense matter can modify the electromagnetic response at an order that is no longer negligible. But this still does not uniquely determine a magnetic permeability law, so a separate EFT matching remains necessary if one wants a robust value of Γ_{struct} .

Second, the field-amplification bookkeeping becomes much more favorable once one distinguishes progenitor surface fields from pre-collapse iron-core fields. For a stellar radius $R_\star \sim 10^{11}$ cm, an iron-core radius $R_{\text{core}} \sim 3 \times 10^8$ cm, and a neutron-star radius $R_{\text{NS}} \sim 1.2 \times 10^6$ cm, the flux-freezing factor from core to neutron star is

$$\Gamma_{\text{flux}} = \left(\frac{R_{\text{core}}}{R_{\text{NS}}} \right)^2 \approx 6.25 \times 10^4.$$

If the progenitor surface field is already in the $10^3 - 10^4$ G corridor, the corresponding pre-collapse core field reaches roughly

$$B_{\text{core}} \sim 1.1 \times 10^8 - 1.1 \times 10^9 \text{ G},$$

which gives a flux-only neutron-star field of order

$$B_{\text{NS}}^{\text{flux}} \sim 6.9 \times 10^{12} - 6.9 \times 10^{13} \text{ G}.$$

In that case the extra structural amplification needed to enter the magnetar regime is no longer enormous. For example, a core seed field $B_{\text{core}} \sim 10^9$ G requires only

$$\Gamma_{\text{struct}} \approx 1.6 \quad \text{for} \quad B_{\text{NS}} = 10^{14} \text{ G},$$

and

$$\Gamma_{\text{struct}} \approx 4.8 \quad \text{for} \quad B_{\text{NS}} = 3 \times 10^{14} \text{ G}.$$

This is the narrowest quantitative statement of the NVG magnetar channel currently supported by the code: if the pre-collapse core field is already high, only a moderate additional amplification is required.

Third, the remnant-energetics argument strongly disfavors obligatory millisecond birth. Using a standard neutron-star moment of inertia, the rotational reservoir is approximately

$$E_{\text{rot}} = \frac{1}{2} I \Omega^2.$$

The script gives

$$E_{\text{rot}}(1 \text{ ms}) \approx 2.8 \times 10^{52} \text{ erg},$$

$$E_{\text{rot}}(5 \text{ ms}) \approx 1.1 \times 10^{51} \text{ erg},$$

$$E_{\text{rot}}(10 \text{ ms}) \approx 2.8 \times 10^{50} \text{ erg}.$$

This sharp drop is precisely why a rotation-independent amplification channel remains astrophysically attractive: it allows strong fields without forcing the system into the most energetically problematic birth regime.

The same script also clarifies the status of the extreme long-period source 1E 161348-5055. A pure vacuum dipole-braking evolution to $P \approx 6.67$ hr within a remnant age of $\sim 2 - 10$ kyr requires an effective dipole field of order

$$B \sim 6.6 \times 10^{17} - 1.5 \times 10^{18} \text{ G},$$

which is far above the ordinary magnetar range. Therefore the NVG structural amplification mechanism may help explain magnetic-field generation without millisecond birth, but it does not by itself explain the most extreme long-period case. For 1E 161348-5055, additional braking physics such as fallback-disk torques, propeller coupling, or another early-time environmental torque is still required.

That additional step has now also been tested in a dedicated script, `verification/nvg_1e161348_fallback_torque.py`. In a simple propeller/fallback-disk model with a decaying accretion rate

$$\dot{M}(t) = \dot{M}_0 \left(1 + \frac{t}{t_0}\right)^{-\alpha},$$

and a standard fastness-dependent disk torque, the target period can be approached within the observed remnant-age corridor using ordinary magnetar-strength dipole fields rather than the unrealistically large fields required by pure vacuum dipole braking. The best-fit benchmark solutions obtained in the current scan lie in the approximate corridor

$$P_0 \sim 0.1 \text{ s}, \quad B \sim 3 \times 10^{14} - 10^{15} \text{ G},$$

$$\dot{M}_0 \sim 3 \times 10^{20} - 10^{22} \text{ g/s}, \quad t_0 \sim 10^4 - 10^5 \text{ s}, \quad \alpha \sim 1.2 - 1.5,$$

yielding final periods

$$P_f \sim 2.45 \times 10^4 - 2.74 \times 10^4 \text{ s}$$

for remnant ages of $2 - 10$ kyr. In other words, 1E 161348-5055 does not force the model into a 10^{18} G dipole field; it forces the model to include disk-mediated braking in addition to field amplification.

Finally, the population-level check has now also been implemented in `verification/nvg_magnetar_population_scan.py` using a live-parsed McGill magnetar catalog subset together with a curated high- B pulsar / transition sample and the long-period outlier. The scan asks a direct question: for the benchmark NVG corridor

$$\Gamma_{\text{struct}} \approx 2.2 - 3.3,$$

what progenitor surface field is required to reproduce the observed dipole field of each source after standard flux freezing? For the working progenitor corridor

$$B_{\star, \text{surf}} \sim 10^3 - 10^4 \text{ G},$$

the result is encouraging but not trivial. In the present catalog-driven scan, 17 of 32 non-outlier objects lie directly inside that corridor. Broken down by class, the benchmark NVG channel already accommodates 10 of 24 live-parsed magnetars, 6 of 7 high- B radio pulsars, and the transition object PSR J1846\$-\$0258, while three objects fall below the corridor and twelve magnetars lie above it. The objects below the corridor are not problematic because they simply do not require a strong structural boost; the objects above it show where the benchmark Γ_{struct} is not sufficient if one insists on the narrowest progenitor-field prior.

An uncertainty layer has now also been added on top of this central estimate. Scanning over a representative grid of progenitor radii, neutron-star radii, and an overall rescaling of the structural-amplification corridor gives a population coverage range of

$$11/32 \text{ to } 24/32,$$

with a central value of 17/32. In this scan only 1/32 objects remain inside the fiducial progenitor-field corridor for **all** tested scenarios, while 31/32 are compatible in **at least one** scenario. This is a useful result because it shows exactly where the present NVG magnetar channel stands: it is already broad enough to intersect most of the observed population somewhere in parameter space, but the detailed coverage fraction is still highly sensitive to the assumed structural-amplification normalization and stellar-scale priors. Thus the current population result is neither a blanket confirmation nor a failure: it shows that the NVG channel already covers a substantial fraction of the observed distribution, while the highest-field magnetars still require either stronger progenitor magnetization, a larger structural amplification at the high-density end, or additional post-birth magnetic evolution.

The scan has now also been promoted one step further into a source-by-source uncertainty-aware summary. Using object-level magnetic-field uncertainties together with the global prior grid, the current sample splits into

$$9 \text{ high-support}, \quad 18 \text{ mixed-support}, \quad 5 \text{ low-support}$$

objects, where “high-support” means that the probability of landing inside the NVG working corridor is at least 0.8. The most stable support currently comes from the transition/high- B pulsar sector and from moderate-field magnetars such as 1E 2259+586, while the weakest support is concentrated in the extreme-field tail, including SGR 1806\$-\$20, *SGR*1900+14, and 1E1841-\$045. This sharpened breakdown is useful because it shows that the tension is not spread uniformly across the catalog: it is concentrated in a specific subset of the highest-field objects.

Finally, the same catalog scan has now been used to construct a simple global pseudo-likelihood for the normalization of the structural amplification itself. When the benchmark interval is rescaled by a global factor and confronted with the catalog, the preferred value shifts upward relative to the original dense-core

benchmark. The current best fit is obtained near

$$\Gamma_{\text{struct}} \approx 5.0 - 7.4,$$

with a narrow support band close to

$$\Gamma_{\text{struct}} \approx 5.0 - 7.6,$$

and a wider tolerated band extending down to approximately

$$\Gamma_{\text{struct}} \approx 2.6 - 7.6.$$

This is an important synthesis result. It says that the original benchmark $\Gamma_{\text{struct}} \sim 2.2 - 3.3$ is already sufficient for a substantial subset of the observed population, but the population as a whole prefers a somewhat larger effective amplification if one wants to cover more of the magnetar tail without invoking extreme progenitor fields for every high- B source.

For practical use in the preprint workflow, the catalog-level breakdown has also been exported into a separate appendix table with one line per source, listing the inferred support probability, the required progenitor surface-field corridor, and the corresponding tension class. This makes it possible to see immediately which objects are stable fits, which sit on the edge of the corridor, and which still demand either stronger amplification or additional braking physics.

6. What Must Be Checked Next

This is the decisive section. The value of the model depends not on how strongly it is stated, but on whether it can survive these checks.

6.1. Derive the Electromagnetic Response from More Standard Microphysics

The first requirement is a better derivation of the effective electromagnetic response in dense matter. At minimum, one needs a transparent bridge between:

1. the NVG/VMF order parameter $\mathcal{W}$ or $M_{\Omega}(n_B)$,
2. standard QCD-matter or many-body response quantities,
3. and the constitutive relation used for magnetic amplification.

This could take the form of a controlled matching calculation, an in-medium polarization analysis, or a more explicit dense-matter EFT in which the same order parameter that modifies the EOS also modifies the gauge sector. Until that bridge is written down more rigorously, the permeability-collapse step should be presented as a motivated hypothesis rather than a finished derivation.

6.2. Check Joint Consistency with the Full Neutron-Star and Supernova Systematics

The second requirement is global consistency. The magnetar mechanism cannot be judged in isolation; it must coexist with the rest of the stellar phenomenology that motivates the paper in the first place.

In practice, I would require the model to reproduce, simultaneously:

1. realistic neutron-star maximum masses and radii,
2. acceptable spin bounds and spin evolution,
3. supernova-remnant energetics consistent with the absence of obvious $\sim 10^{52}$ erg rotational injection in the problematic cases,
4. and a sensible distribution of magnetic-field strengths across ordinary pulsars and magnetars.

If the same parameter choice that boosts magnetic fields destroys the EOS, overproduces spin-down power, or conflicts with remnant energetics, the mechanism fails. Conversely, if one parameter region can satisfy all of these constraints together, the proposal becomes much stronger than a one-observable argument.

6.3. Compare Against the Magnetar Population, Not Just the Paradox Story

The third requirement is a population-level test. A convincing claim is not merely that one paradox exists, but that the proposed mechanism predicts the right class of objects. A first catalog-driven population scan with an explicit uncertainty layer has now been completed, but it should still be extended into a fuller end-to-end catalog fit with explicit selection effects and source-by-source uncertainties.

The immediate targets should include:

1. long-period magnetar candidates such as 1E 161348-5055,
2. magnetars with inferred low or moderate birth spins,
3. the broader Galactic magnetar sample,
4. and the transition region between high-field radio pulsars and magnetars.

The key question is whether the model predicts a realistic occupancy of parameter space: field strength, present spin period, spin-down rate, age, and remnant environment. If the claim is “field amplification without millisecond birth,” then it should be confronted directly with exactly those systems for which millisecond birth is least plausible.

7. Reframed Interpretation of the Magnetar Claim

The strongest defensible statement at present is not that NVG has already proved a large vacuum-permeability collapse in neutron-star cores. It is this:

There exists within the NVG/VMF dense-matter program a concrete and testable route to a rotation-independent magnetar channel, provided that the electromagnetic response of the same structural sector can be derived with sufficient rigor and shown to remain compatible with neutron-star and supernova observations.

That is already a worthwhile claim. It narrows the problem, defines the missing derivation, and turns the magnetar paper from a qualitative alternative to the dynamo scenario into a falsifiable research program.

8. Conclusion

The slow-rotating magnetar paradox remains a legitimate challenge to the strongest version of the millisecond-dynamo picture. The NVG/VMF framework offers a plausible structural alternative: dense-matter phase evolution may amplify magnetic fields without requiring extreme birth rotation. The new quantitative closure checks strengthen that statement in one specific sense: for pre-collapse core fields already in the $10^8 - 10^9$ G corridor, only a modest additional amplification is needed to reach ordinary magnetar fields, while the rotational-energy budget no longer forces the system into the $\sim 10^{52}$ erg millisecond-birth regime. But the present status of the idea is still mixed.

The dense-matter backbone of the model is comparatively developed, with EOS and neutron-star benchmarks already embedded in the project. The electromagnetic bridge is only partial. For that reason, the correct scientific posture is not to overclaim a completed derivation, but to state clearly what would validate or falsify the scenario.

If a standard-microphysics derivation of the effective magnetic response can be built, and if the same framework simultaneously explains neutron-star masses, spin limits, remnant energetics, and the broader magnetar population, then the case for a rotation-independent magnetar channel becomes serious. The present calculations already show four sharper boundary conditions than before: first, 1E 161348-5055 remains too extreme for dipole braking plus field amplification alone, but it becomes quantitatively accessible once a fallback-disk propeller torque is included; second, the population-level success fraction is currently prior-sensitive rather than rigid; third, the main residual tension is concentrated in the highest-field magnetar tail rather than in the transition/high- B pulsar sector; fourth, the catalog itself prefers an effective amplification somewhat larger than the original dense-core benchmark, with a best-fit corridor near $\Gamma_{\text{struct}} \sim 5 - 7.4$. That is the correct narrowed claim. If the framework cannot

satisfy these combined demands, the mechanism should be rejected. That is exactly the standard this version is designed to make explicit.

Data Availability

The theoretical background, verification scripts, and supporting EOS calculations used to motivate this version are contained in the NVG repository, including the dense-matter EOS, joint neutron-star inference, electromagnetic formalization notes, the magnetar closure script `verification/nvg_magnetar_closure.py`, the fallback-disk torque scan `verification/nvg_1e161348_fallback_torque.py`, and the catalog-driven population scan `verification/nvg_magnetar_population_scan.py`: [<https://github.com/infosave2007/vmf>] and Zenodo [<https://zenodo.org/records/20214457>].

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